

Coupling-gated cortical coherence: a conditional framework for unity and self-representation

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19 September 2026

Abstract

How could activity distributed across cortex support one experienced situation? We separate three requirements and state the conditions that connect them: *coherence*, meaning collective phase order; *compatibility*, meaning that local descriptions agree where their domains overlap; and *self-representation*, meaning that the resulting state carries an accurate reconstruction of itself. Four results carry the argument. Installed energy bounds the mean coupling a declared actuator supplies, so a coherent stationary branch requires it to exceed $2D/\kappa$, with D the phase diffusion and κ set by the mode prices; energy is not the scarce side of that condition for a macroscopic substrate, which leaves it a statement about κ , derived here for neither cortex nor a processor. Phase encoders shared on overlaps turn phase order into a bound on content disagreement, vacuous for a whole population and informative within a patch. Compatible local states glue to a unique global state. A contracting reconstruction that fixes that state makes it the unique fixed point, while accuracy across a family of states requires distinguishable encodings. Lean 4 checks these results and their composition, with the eight connecting hypotheses

as explicit arguments and no axioms beyond its libraries’. Numerical controls bound what observation can decide: distinct mechanisms share stationary phase distributions, finite ramps shift the apparent onset exponent, a winding field and a uniform one are stationary at one coupling with resultants spanning nearly the range of that observable, and a coupling rule descending a squared-drift objective holds coherence while losing the structure it was meant to acquire. We propose the glued state and its internal reconstruction as correlates of unity and a minimal self, and an extracellular-field hypothesis linking awakening to rising coupling. Testing it requires calibrated geometry, resolved phase dynamics, validated decoders and a spatial observable beside the resultant.

1 Introduction

How could activity distributed across cortex support one experience? A field account proposes that endogenous extracellular electromagnetic fields partly coordinate spatially separated neural processes (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Successful coordination would still leave a representational question: what makes the coordinated activity a description of one situation, with a perspective on that situation?

We distinguish three requirements. *Coherence* concerns collective phase order. *Compatibility* concerns whether local descriptions agree where their domains overlap. *Self-representation* concerns whether the resulting state carries an accurate internal reconstruction of itself. We propose the latter two relations as formal correlates of conscious unity and a minimal self; their

15 interpretation requires evidence beyond the phase model.

16 Unity has been approached differently elsewhere: integrated information
17 theory treats integration and exclusion as constitutive (Tononi et al., 2016),
18 global-workspace accounts tie availability to broadcast (Baars, 1997; Dehaene
19 et al., 2014), and predictive-processing accounts derive both from a hierar-
20 chical generative model (Friston, 2010). Here the three requirements are
21 kept apart rather than derived from one another, so that each carries its own
22 assumptions and admits its own measurement; Section 8.4 returns to the
23 comparison.

24 Consider reaching for a cup. A visual description of its location and a
25 description of the hand’s relation to it can agree about where the cup is while
26 contributing different contents. Synchronizing the two processes would leave
27 disagreement about that shared quantity unresolved. An internal representa-
28 tion must additionally locate the perceiver within the represented situation.
29 This example motivates the requirements; the mathematical constructions
30 use spatial measures and fields in real normed vector spaces, whose identifi-
31 cation with perceptual content is a further modelling task.

32 The mathematical contribution is to specify connections between these
33 requirements and the assumptions each connection needs. A declared ac-
34 tuator’s stored energy bounds its mean coupling. Phase encoders shared on
35 overlaps convert phase order into a bound on content disagreement. Compat-
36 ible spatial measures have a unique global extension; approximate agreement
37 controls a specified global selection. A contracting reconstruction makes that

extension its unique fixed point only if it accurately reconstructs the extension. Accuracy across a family of states imposes a further condition on the information retained by the encoding.

The empirical contribution is to identify what observations can decide. Stationary phase distributions can be shared by stronger coupling, weaker diffusion and a matched common drive. Finite ramps change the observable onset. A winding field is stationary at the same coupling as a uniform one and has the opposite resultant, so that observable does not determine the state. Coherent adaptation can lose alignment with a designated environmental template, and a compatibility estimator needs diagnostics that distinguish content from decoder artifacts. These findings determine the measurements needed to test the proposed connections.

Three of the ingredients are standard: the synchronization threshold $K_c = 2D$ for the noisy mean-field model (Sakaguchi, 1988), the unique gluing of compatible local sections, and the Banach fixed-point theorem (Banach, 1922). Four things here are not. The installed-energy condition bounds mean coupling by the stored energy of a declared actuator, which makes a resource condition necessary for coherence on the stationary branch. The coherence-to-content bound converts phase order into a statement about descriptions rather than about timing, and its patch-local form is what keeps it from being vacuous in a large population. A stability criterion on the reweighted coupling kernel, together with an existence condition on an amplitude layer, settles when a low resultant reports weak coupling and when

61 it reports a winding field. And the composition is machine-checked together
62 with a witness discharging all eight hypotheses at once, which is what makes
63 their conjunction non-empty rather than a list no object satisfies.

64 Granting those premises, collected in Table 1, the account runs as follows.
65 Extracellular contraction on awakening raises endogenous field amplitude
66 and with it the effective coupling between cortical oscillators. Once that
67 coupling passes $2D$, a coherent phase branch exists, and from an initial spread
68 within a quarter turn the population settles onto it; other initial data leave
69 windings stationary at the same coupling. Encoders shared where cortical
70 territories overlap turn that phase order into agreement about the quantities
71 the territories describe in common, so the local descriptions glue: there is
72 one state of which each is a restriction. Part of that state encodes the state
73 itself, and a readout of that part reconstructs it accurately enough, and
74 sensitively enough across the states that matter, to make it the unique state
75 the system models itself as being. On this account the glued state is what
76 makes distributed activity one situation, and its accurate self-reconstruction
77 is what gives that situation a perspective. Every step of it is conditional.

78 Sections 2–4 develop the three requirements together (Fig. 1). Section 5
79 composes them, stating the thermodynamic constraints and all eight con-
80 necting hypotheses per branch; its supporting models of feedback, learning
81 and resource supply have distinct state spaces and assumptions. Section 6
82 then examines what the observables can establish, before Section 7 devel-
83 ops the cortical hypothesis and an awakening experiment. The hypothesis is

84 that extracellular contraction increases field-mediated coupling; its direction,
 85 rate and relation to recovery require independent calibration. Supplemen-
 86 tal Material organizes the constructions and controls behind the claims they
 87 support.

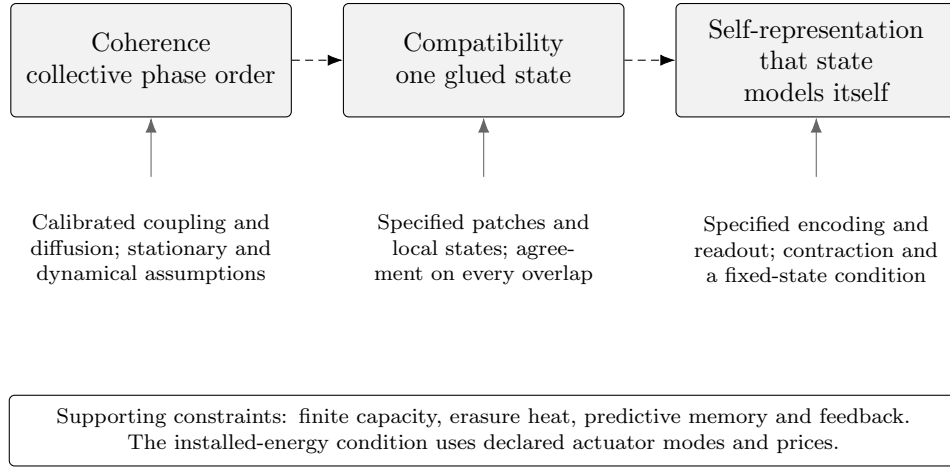


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

88 2 Coherence: physical coupling and its re- 89 source condition

90 2.1 Dynamics and the stationary threshold

91 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

92 Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and
93 K_{ij} has inverse-time units, with angles dimensionless. Uniform mean-field
94 coupling is $K_{ij} = K/N$; frequency heterogeneity and diffusion are distinct
95 parameters. The stationary calculation below uses identical frequencies and
96 this scalar coupling. Applying it to a spatial kernel requires an aggregation
97 rule and a check of the resulting approximation.

98 For identical frequencies in their rotating frame, the positive normalized
99 stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

100 where I_n are modified Bessel functions and ψ is the collective phase. The for-
101 mal derivation uses the absence of stationary current for a periodic gradient
102 drift to obtain this density, then verifies normalization and self-consistency.
103 At $K \leq K_c = 2D$, only $r = 0$ is self-consistent (Sakaguchi, 1988). Above

threshold, exactly one positive solution coexists with zero and increases
strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

Figure 2 shows this stationary branch.

Linearization about the uniform density gives rate $K/2 - D$ for the first harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists exactly above threshold. About the coherent density, the quadratic form is strictly dissipative transverse to collective rotation, whose rate is zero. These linear estimates concern the limiting equation. Relating them to finite stochastic trajectories requires a dynamical mean-field limit and branch selection; exchange symmetry alone does not establish the needed independence (Sznitman, 1991).

2.2 Installed energy bounds mean coupling

A spatial model replaces the finite sum by integration against a kernel. One declared implementation carries response modes with continuous profiles ϕ_i , installation prices u_i and nonnegative occupancies $c_i(z)$ in configuration z :

$$K_z(x, y) = \sum_i c_i(z) \phi_i(x) \phi_i(y), \quad U(z) = \sum_i u_i c_i(z). \quad (4)$$

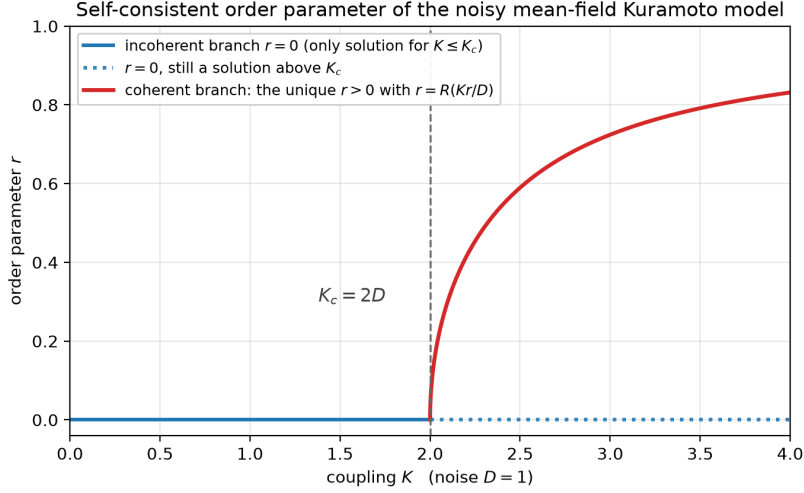


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

119 The same configuration sets the reciprocal kernel and stored energy. An
 120 executed transition changes the kernel through its changed occupancies and
 121 requires work $U(z') - U(z) + Q$, with Q positive into the reservoir. The
 122 installed kernel is averaged over the resulting configuration law $p(z)$. Cell-
 123 pair reconstruction on refining spatial partitions converges to its product-
 124 measure integral under the specified compactness and sampling assumptions;
 125 the refinement index is independent of the number of control updates.

126 Normalize the substrate measure μ to one. Write $m_i = \int \phi_i d\mu$ and
 127 declare $\kappa > 0$ with $m_i^2 \leq \kappa u_i$ for every mode. Factorizing each mode's

128 integral gives

$$\begin{aligned}
U_{\text{inst}} &= \sum_z p(z) U(z), \\
K_{\text{mf}} &= \iint K(x, y) d\mu(x) d\mu(y) = \sum_z p(z) \sum_i c_i(z) m_i^2 \leq \kappa U_{\text{inst}}.
\end{aligned} \tag{5}$$

129 If this mean coupling is the K of the scalar model, $\kappa U_{\text{inst}} \leq 2D$ excludes a
130 coherent stationary branch and any growing Fourier mode about the uniform
131 density. Thus coherence on that branch requires $U_{\text{inst}} > 2D/\kappa$. The condition
132 is necessary: energy above it need not produce sufficient coupling because
133 the bound can be loose.

134 The quantity constrained is installed energy. Dissipated heat does not
135 determine it, and κ is supplied by the mode profiles, prices and substrate
136 normalization. Applying the scalar threshold to heterogeneous or signed spa-
137 tial dynamics requires a justified reduction. Identifying the installed kernel
138 with a cortical field is a separate physical commitment.

139 **3 Compatibility: from coordinated activity** 140 **to a global description**

141 **3.1 Why common timing is insufficient**

142 For overlapping patches U and V , compatibility means that restricting either
143 local description to $U \cap V$ gives the same object. Descriptions can agree on

144 shared quantities while contributing different information elsewhere. Com-
 145 patible sections of the specified sheaf have a unique global extension. On
 146 a finite cover, local finite spatial measures agreeing as measures on every
 147 overlap also have a unique finite-measure extension whose sheaf image is the
 148 glued section. No mass-one normalization is imposed.

149 In the cup example, visual and reaching descriptions must express a
 150 shared quantity, such as cup location, in common coordinates. Restriction
 151 retains that quantity while dropping details particular to each description.
 152 Agreement permits surface appearance and hand configuration to contribute
 153 to one situation. The mathematical counterexample assigns the same phase
 154 to every patch while two local mass profiles disagree on a shared site. Those
 155 profiles have no common extension. Because they are chosen independently
 156 of phase, synchrony places no constraint on their disagreement.

157 This distinction limits the biological interpretation. Receptive territories
 158 and content domains must be related to spatial patches, and the descriptions
 159 must support the required extension. Matching overlap marginals of de-
 160 coded probability laws need not determine a unique global joint distribution
 161 (Abramsky & Brandenburger, 2011).

162 **3.2 When coherence does constrain content**

163 A declared relation between phase and content gives a positive result. Let
 164 contents take values in a real normed vector space E . Patch i reads an
 165 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point

166 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance and
 167 require $e_{i,x} = e_{j,x}$ wherever patches i and j overlap. Encoders may vary across
 168 sites; their equality on overlaps is a supplied relation between descriptions of
 169 the shared quantity. The order-parameter identity $N^2 r^2 = \sum_{ij} \cos(\theta_i - \theta_j)$
 170 implies $1 - \cos(\theta_i - \theta_j) \leq N^2(1 - r^2)$ for each pair. Consequently every
 171 overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2} N\sqrt{1 - r^2}. \quad (6)$$

172 Here discrepancy is the norm of the difference. At $r = 1$, the observa-
 173 tions agree exactly. The factor N makes this estimate potentially weak
 174 in large populations; it supplies a pointwise guarantee from a population
 175 average. The population in it need not be the whole array. A patch car-
 176 ries an order parameter of its own, and the same bound read there car-
 177 ries that patch's resultant and that patch's site count (`chord_le_of_patch_`
 178 `coherence, compatible_of_patch_coherence`), so a large population ob-
 179 served through small patches is constrained by the small count. A shared
 180 scalar encoder is a special case. Dropping the overlap relation permits dis-
 181 agreement even at perfect locking. A common nonexpansive linear projection
 182 preserves agreement, but agreement of a projection does not establish agree-
 183 ment of the full vector.

184 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
 185 ηo_i . Content and observation discrepancies bounded by ε and δ give a next

186 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
 187 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (7)$$

188 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-
 189 duce an initial mismatch. Approximate observations leave a nondecaying
 190 term in the error bound. Taking a uniform bound on Eq. (6) connects this
 191 update to the phase process.

192 **3.3 What approximate agreement determines**

193 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
 194 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
 195 Weights that vanish outside their patches and sum to one at each site select
 196 a global profile by weighted averaging. The selection lies within ε of each
 197 patch there, and any two such selections differ by at most ε , independently
 198 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
 199 $L\varepsilon$ across selections.

200 At zero tolerance, the selection is the unique exact extension. At every
 201 positive tolerance there are families with no exact extension; selection then
 202 gives an approximation whose dependence on weights is bounded. Supple-
 203 mental Material gives a three-site witness attaining the bound. The mass-
 204 profile construction uses the scalar case of the content dynamics and addi-
 205 tionally requires nonnegativity of the initial profiles and observations. The

206 general vector result supplies no identification with the spatial-measure sheaf.

207 These constructions concern declared objects: a specified cover, contents
208 valued in a normed space, and encoders related on overlaps. They select
209 neither the neural cover nor the decoder. Identifying their contents with
210 perceptual content, and establishing that the identification bears on experi-
211 ence, is a further modelling task; Section 6 sets out what a measurement of
212 it would have to establish.

213 4 Self-representation: a state that models it- 214 self

215 4.1 The reflexive construction

216 Compatibility gives a global state. Reflexivity relates it to an encoding within
217 that state. Equip global sections with a complete metric. If the additional
218 restriction-resonance condition identifies encoding with restriction to a region
219 U , the reconstruction map is

$$F(s) = \text{readout}(s|_U). \tag{8}$$

220 The region and readout are fixed across the state space. If F contracts and
221 fixes the section s_* formed from the local descriptions, Banach's theorem
222 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89

223 supplies the contraction and $F(s_*) = s_*$; restriction resonance is separate.
 224 Supercritical coupling makes the stipulated factor $q = e^{-(K-2D)\tau/2}$ less than
 225 one for $\tau > 0$. Constructing a map with that factor remains a modelling
 226 obligation.

227 In the cup example, the global state includes the represented scene and the
 228 perceiver's relation to it. A local encoding must retain enough information
 229 for the readout to recover that modelled state; cup position alone would be
 230 insufficient if hand configuration and viewpoint could not be reconstructed.

231 For the same map with Lipschitz factor $q < 1$, approximate reconstruction
 232 gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - q}. \quad (9)$$

233 At the stipulated rate this radius grows as $2\varepsilon/((K-2D)\tau)$ near threshold, for
 234 fixed positive ε and τ . Exact uniqueness survives while the error guarantee
 235 weakens.

236 4.2 Reconstruction across a family of states

237 Sensitivity can be required quantitatively. Declare relevant macrostates, an
 238 encoder E and readout R , and require $d(s, R(E(s))) \leq \varepsilon$ throughout that
 239 family. If two states have the same code, the triangle inequality gives

$$E(s) = E(t) \implies \max\{d(s, R(E(s))), d(t, R(E(t)))\} \geq \frac{1}{2}d(s, t). \quad (10)$$

240 Thus M relevant states pairwise more than 2ε apart require at least M
241 distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$.
242 Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the
243 existing self-prediction map.

244 This is an additional accuracy criterion on a family, with no requirement
245 for multiple exact fixed points. A blind encoder or constant readout fails it
246 for separated states despite having a unique fixed point.

247 Where the encoding is read as the state's own restriction to a region, the
248 criterion becomes a condition on the region. Restriction resonance requires
249 the encoder and that restriction to identify the same pairs of states, so a
250 boundary reconstructing two relevant states more than 2ε apart while its re-
251 gion reads them identically is not resonant (`not_isRestrictionResonance_`
252 `of_reconstructs`); read forwards, a resonant boundary that reconstructs
253 a pairwise separated family restricts injectively on it (`restrictToAvatar_`
254 `injOn_of_isRestrictionResonance`). Which states a region reads identi-
255 cally is fixed by where the region sits, before an encoding is chosen, so the
256 resolution the family demands is the region's and no readout supplies it. The
257 implication runs one way: a region that does resolve the family is not thereby
258 shown to carry a resonant encoding, and E89 supplies no such resolution.

259 4.3 Agreement after a change takes communication

260 Snapshot compatibility does not specify how quickly descriptions respond
261 when a shared quantity changes. In a synchronous network whose updates

262 read only the site’s own state and declared neighbours’ messages, two initial
 263 configurations agreeing on a site’s causal past through T rounds produce the
 264 same state there at round T . If two fresh interventions outside that past
 265 require reports separated by more than 2ε , no report from that site can be
 266 within ε of both. This follows from the reconstruction bound in Section 4.2.

267 A three-site delay line witnesses failure after one round and exact report-
 268 ing after two; cutting the path prevents delivery at every round. A report
 269 may still happen to be correct about one fixed world. The bound concerns
 270 responsiveness across interventions.

271 Throughout this section *prediction* names a reconstruction map; reading
 272 it as temporal neural prediction would require dynamics. The relevant family
 273 of states, the metric, the tolerance, the communication graph and the dead-
 274 line are all declared inputs, as are the represented variables and the fidelity
 275 counted as accurate. Reading the resulting code count as physical memory
 276 capacity requires a hardware model, and a readout that varies with the state
 277 requires a further one.

278 5 Thermodynamics and composition

279 The phase and representation results concern coupling, content disagreement
 280 and reconstruction error. Thermodynamics constrains the physical opera-
 281 tions that could implement them. Finite-register capacity obeys $H(p) \leq$
 282 $\log |X|$; erasure heat depends on specified system–bath assumptions (Lan-

283 dauer, 1961; Norton, 2005; Shenker, 2000). A separate boundary result con-
 284 cerns a field reaching disconnected vacuum components (Kibble, 1976). The
 285 composition connects these objects through the explicit hypotheses in Ta-
 286 ble 1.

287 5.1 Feedback changes the information constraint

288 For passive memory X_t driven by an environmental signal S_t , the predictive
 289 bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (11)$$

290 The Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$ makes the bracket nonnegative (Still
 291 et al., 2012). Zero retained information also satisfies it, so a task and a
 292 learning rule require additional specification. Relating this bound to Lan-
 293 dauer heat requires a common thermal scale and budget, and memory waste
 294 bounded by erased entropy.

295 An acting agent changes the signal it will receive. Specify policy, world,
 296 observation and memory-update channels. For their finite laws,

$$\begin{aligned} I(X_t; S_t) - I(X_t; S_{t+1}) &= I(X_t; S_t \mid S_{t+1}) \\ &\quad - I(X_t; S_{t+1} \mid S_t). \end{aligned} \quad (12)$$

297 The second conditional term vanishes in the passive model. With feedback,
 298 the signed difference can be negative as action creates correlations with the
 299 future. A causal reading requires an environmental state sufficient to describe

300 the relevant dynamics.

301 For a controlled substep, the active thermodynamic bound instead uses
302 the joint entropy. Forward and reverse path laws give $\Sigma = D_{KL}(P_F \| P_R) =$
303 $\Delta H(X, S) + \langle Q \rangle / \theta \geq 0$ under local detailed balance, with $\theta = k_B T > 0$ and
304 heat positive into the reservoir (Horowitz and Esposito, 2014; Ito and Sagawa,
305 2013). Every forward path of positive probability must have a positive-
306 probability reverse for this finite identity. A missing reverse gives infinite
307 divergence. The supplement develops the support conditions and composes
308 actuation with memory updates on their shared intermediate law.

309 **5.2 Learning and repeated operation need explicit re-** 310 **sources**

311 The finite examples distinguish selecting a feasible policy, relaxing a register
312 toward an encoded reward, and learning an unknown task from observa-
313 tions. The observational learner acquires correlation with an environmental
314 parameter while its action marginal stays uniform. A continuing model exe-
315 cutes actuation, observation-driven learning and reset in sequence, preserving
316 above-chance performance at a positive work cost per cycle.

317 A phase-derived sensor makes the informational distinction concrete: an
318 informative readout and a blind control have the same phase order but dif-
319 ferent learning performance. Attaching the learner’s register to the actuator
320 of Section 2.2 ties reward, installed kernel and work to its own executed law.

321 The learning witness rises above the necessary installed-energy bound while
 322 its matched control remains on the exclusion line. These finite constructions,
 323 detailed in Supplemental Material, leave the evolving feedback between phase
 324 dynamics, installation and sensing to be specified.

325 For that model, common energy E , thermal scale θ , local detailed balance
 326 and positive support at each stage give

$$\theta[H(p_0) - H(p_N)] \leq Q_{<N} = W_{<N} + \langle E \rangle_0 - \langle E \rangle_N \leq B + \langle E \rangle_0 - \langle E \rangle_N, \quad (13)$$

327 where expected cumulative work stays within the supplied allowance B at
 328 every prefix. This bounds expectations. Separate constructions track a store
 329 on each trajectory, include finite sources, charge sensor clearing and prepend
 330 preparation. They specify when a mean allowance represents an available
 331 resource; an expected budget alone permits overdrawn paths.

332 These constructions establish costs for their declared processes. A finite
 333 source with a positive net cost permits only finitely many funded cycles,
 334 while continuing operation requires replenishment. Register–bath witnesses
 335 also distinguish actual bath-energy transfer from transition-log heat. Identifying a work store, an erasure budget and the installation energy of Section 2.2 requires physical relations that these separate models do not supply
 336 automatically.
 337
 338

339 5.3 What the composition establishes

340 The passive theorem `chain` and feedback theorem `chain_active` each ac-
 341 cept eight connecting implications (Table 1). In the active branch, a named
 342 substep at the register’s thermal scale satisfies

$$\theta > 0, \quad \theta[H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (14)$$

343 with B the register’s heat. The upper allocation follows from a ledger whose
 344 operations deliver that total heat to one reservoir and whose other operations
 345 do not increase joint entropy. Those ledger assumptions are physical inputs.
 346 Neither Landauer’s lower bound nor microscopic reversibility supplies them.

Table 1: Connecting hypotheses, eight per branch, and the installed-energy consequence. The branches share E12, E23 and E56–E89. Exact overlap compatibility and the encoding–readout structure are additional inputs. Supplemental Material, Table S1, maps the node-level results.

Connection	Status	Required assumption
E12 <i>capacity and boundary</i>	Physical premise	A finite-capacity system realizes a field leaving its vacuum manifold. Capacity and boundary are independent roots.
E23 <i>boundary to register</i>	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.
E34 / E34Active <i>register to dissipation</i>	Bridge assumption	Passive: predictive memory at the register’s thermal scale and heat budget, with nonzero waste. Active: positive paths, local detailed balance and allocation of register heat to the named substep; the allocation uses the ledger identity and other operations’ compressiveness.

Table 1 (continued)

Connection	Status	Required assumption
E45 / E45Active <i>couplings to continuum</i>	Modelling assumption	Convergent spatial coupling energies. Declared mode hardware supplies a kernel limit for either branch. A learning witness computes installation and work from its own executed law. Refinement and physical identifications remain inputs.
E56 <i>field identification</i>	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67 <i>above threshold</i>	Physical commitment	That field operates above the synchronization threshold.
E78 <i>coherence to cover</i>	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements.
E89 <i>contraction to fixed point</i>	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $q < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.
Installed energy	Conditional theorem	Under Section 2.2’s occupancy and mode-price conditions, E56 and E67 require $U_{\text{inst}} > 2D/\kappa$. The learning witness clears this necessary bound; its matched control does not. Neither physical premise follows.

Both branches also take a thermodynamic cover carrying exact over-
lap compatibility and a reflexive boundary specifying encoding and readout.
They conclude **UnifiedSelf**: the glued section is the unique fixed point of
self-prediction. Compatibility and accuracy at that section enter as premises;
uniqueness follows. Identifying encoding with restriction needs the separate

352 resonance condition of Eq. (8).

353 Each of the eight edges is supplied rather than derived from the one above
354 it. E12 is equivalent to its conclusion, so capacity and boundary are inde-
355 pendent roots rather than a step from one to the other; both E45 discharges
356 take their convergence from refinement data and not from the thermody-
357 namic bound; E78 states that coherence relaxes the supplied cover, and the
358 coherence-to-content estimate of Eq. (6) bounds disagreement only for en-
359 coders already shared on overlaps. The inferential content of the composition
360 is its last step, where contraction and a fixed glued section give uniqueness.
361 What composing them establishes is that the premises are consistent, named
362 in one place and machine-checked together, and not that coherence follows
363 from installed energy or self-representation from coherence.

364 Joint witnesses combine a register, boundary, refining mesh and small
365 spatial cover; active witnesses add controlled thermal updates and a priced
366 installation. They establish satisfiability under their stated identifications.
367 Mesh convergence follows from refinement data, and the witnesses establish
368 no common cortical mechanism. Supplemental Material maps the claims to
369 their Lean declarations and records the counterexamples. The development
370 declares no additional axioms beyond its libraries' foundational axioms.

371 6 What the observations establish

372 The composition holds under its premises. Whether observation can establish
373 those premises is a separate question, and the three requirements it connects
374 concern different objects: phase dynamics, local descriptions and internal
375 encodings. Their observables must retain these distinctions. The following
376 controls show which mechanisms remain indistinguishable, which temporal
377 approximations fail, which states one observable cannot separate, and what
378 independent content measurements must establish.

379 6.1 Stationary distributions leave the mechanism un- 380 determined

381 Equation (2) gives a concentration–coherence relation with no fitted shape
382 parameter. Estimate r from circular resultant length and a from density
383 shape: inverting R from the measured r would make agreement automatic.
384 Even independently estimated agreement identifies a statistical family shared
385 by different mechanisms.

386 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
387 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
388 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
389 oscillators with prescribed drift $-h(t)\sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,
390 and diffusion D_0 have that same density. Their stationary distributions co-
391 incide window by window despite different causes.

Resolved dynamics supply additional information. In the ideal model with continuous paths, quadratic variation accumulates at rate $2D$ and conditional increment means constrain drift. Separating endogenous coupling from a matched common drive additionally requires a measured drive or controlled perturbation; identifying electromagnetic coupling requires evidence specific to that interaction. Supplemental Material gives the equivalence construction and its numerical check.

6.2 Finite ramps change the observable onset

The square-root law in Eq. (3) describes stationary solutions. A changing coupling requires time for the phase distribution to approach those solutions. In a deterministic mean-field control that crosses the threshold symmetrically, none of the scanned rates meets the declared tracking tolerance: slowing the crossing does not remove the terminal residual under that initialization. A slow control variable therefore supplies no general guarantee of branch tracking at onset. Supplemental Material gives the residual, rate range and admissible rates away from threshold.

The finite-population ramp study uses $N = 2000$, $D = 1$ and 32 replicas at 8 coupling speeds from $v = 0.1$ to $v = 0.0001$. Escape is the first of three consecutive saved samples at $r \geq 0.2$ after threshold. Faster ramps delay escape (Fig. 3); only 5/32 replicas escape within the fastest window. The 6 uncensored legs supply the delay scaling below; the decade-spaced ones among them have mean coupling delays 0.1947, 0.0684 and 0.0253. Fitted

onset exponents differ from those of the exact stationary branch evaluated on the same grids. At the fastest speed the ensemble mean never reaches the fit window, so its fitted slope does not estimate stationary onset.

Delayed passage through a noisy bifurcation (Berglund & Gentz, 2002) makes the apparent exponent depend on ramp speed and fitting window. The specified deviation-based critical ramp speed is not bracketed by this sweep. With fluctuating coupling, the relevant stationary comparison also depends on rate: the mean coupling approximates order in the tested fast regime, while averaging stationary order over coupling works for the tested slow drives confined to one side of threshold. Slow crossing drives can satisfy neither approximation. These are finite-horizon observations, with controls and parameters in Supplemental Material.

The delay scaling is the prediction this section supports, and it needs no absolute calibration. Over the 6 uncensored legs the mean delay follows $\Delta K \propto v^{0.444}$, with a 95% slope interval of $[0.394, 0.494]$. The reference for that exponent is the same estimator applied to a deterministic mean-field run seeded at the fluctuation floor, which returns 0.425 over the same legs at the same escape criterion, inside the interval on the measurement. The distance from $1/2$ is what an absolute escape criterion costs: the run crosses the saturation between the floor it starts from and the level it is stopped at, and tightening the criterion to $r \geq 0.05$ returns 0.490. Stated as a law rather than as a number: onset delay scales as the square root of the rate at which coupling crosses threshold. Testing it requires only that $\dot{K}$ be monotonic

437 in a manipulated rate — not its value, nor κ , nor γ — so it is a within-
 438 subject design that varies the rate of emergence from anaesthesia alone, with
 439 each participant as their own control. Such a design must avoid the hazard
 440 this sweep exhibits: a fixed absolute escape criterion meets a size-dependent
 441 critical fluctuation floor, so the criterion has to be referred to each recording's
 442 own pre-threshold fluctuation scale rather than to a shared number. The
 443 diffusion D is the remaining input, and Section 6.1 notes that it is estimable
 444 from resolved phase traces through the rate $2D$ at which quadratic variation
 445 accumulates, which is a different quantity from the concentration the propofol
 446 exercise could not range over.

447 Additional controls address integration bias, population size and spatial
 448 or signed coupling. The refined signed-network crossing bracket $K_{\text{eff}}/D \in$
 449 $[1.958, 2.009]$ contains 2. These are finite numerical regimes, with full pa-
 450 rameters and controls in Supplemental Material; they supply no cortical cal-
 451 ibration.

452 **6.3 A winding state leaves the order parameter am-** 453 **biguous**

454 The order parameter r is indifferent to where a phase sits in space. A field
 455 winding once across a periodic sheet therefore has $r = 0$ while every patch
 456 of it stays locked. Such a field is not an artificial construction here. On a
 457 torus a winding is stationary for any isotropic symmetric kernel at identical

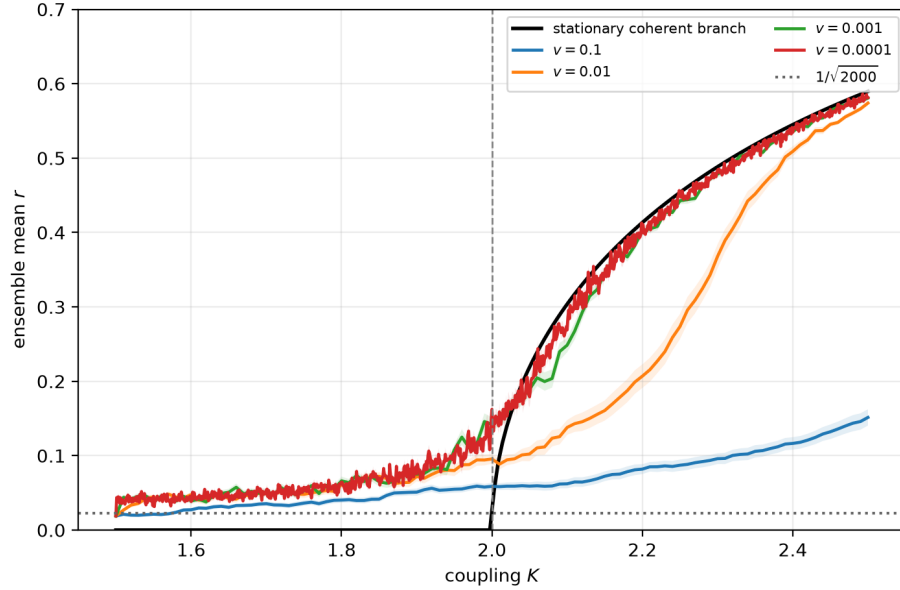


Figure 3: Finite-rate departures from the stationary branch. Ensemble-mean order versus coupling for four ramp speeds v , with one-standard-error bands across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape, changing the apparent recovery onset.

458 frequencies: the coupling term is odd in the separation against a kernel even
 459 in it, so the drift cancels pair by pair. `char_is_kuramoto_trajectory` proves
 460 this for any finite periodic array, this sheet included. A winding satisfies the
 461 positivity and identical-frequency hypotheses. It violates only the quarter-
 462 turn initial spread that the convergence result additionally requires before
 463 placing the limit at the uniform state.

464 On the sheet of Section 2's spatial control, a winding survives below
 465 an interpolated decay length of 0.1871 mm and unwinds above it. That
 466 boundary is interior to the 0.1–0.3 mm band. At 0.1 mm the same sheet at the

467 same coupling, diffusion and kernel reaches $r = 0.9657$ from a uniform start
 468 and $r = 0.0083$ from a winding one. Both are stationary, with patch-local
 469 order 0.9015 and no winding defect. At fixed K the observable therefore takes
 470 two values spanning nearly its whole range. Coupling does not determine the
 471 resultant, and a low resultant does not by itself distinguish weak coupling
 472 from a winding field.

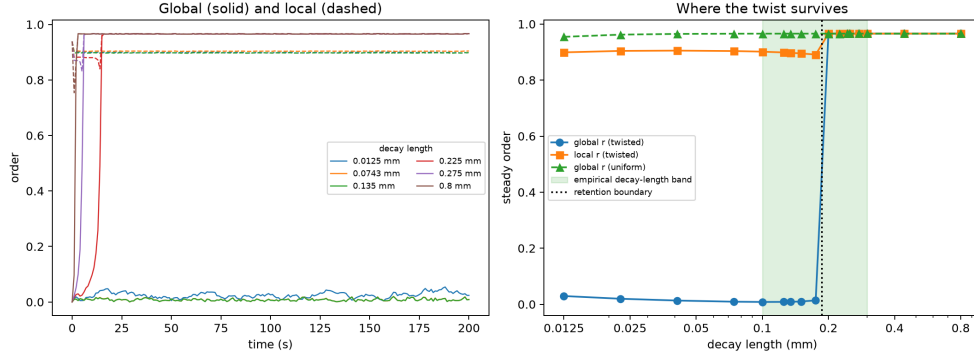


Figure 4: A winding phase field on the spatial sheet, at identical frequencies. Left: global order (solid) against patch-local order (dashed) across a subset of coupling decay lengths. Right: the twisted and uniform initial conditions compared across lengths, with the shaded 0.1–0.3 mm band and the interpolated retention boundary. Below that boundary the two initial conditions reach different stationary values of r at the same coupling, diffusion and kernel, so the resultant does not report the coupling. The sweep is sampled at the lengths marked; it establishes no transition and no biological identification.

473 Under the quenched frequency spread of 1.5 rad s^{-1} that the spatial con-
 474 trol carries, the boundary moves to 0.1421 mm. The same comparison at
 475 0.1 mm gives 0.9418 against 0.0348, holding across 3 frozen realizations.
 476 Heterogeneity shortens the range that sustains a wave without removing it.

477 Which windings persist has a criterion behind it. Linearizing the phase
 478 model about a winding returns a coupling of the same translation-invariant
 479 form, each separation reweighted by the cosine of the phase the winding
 480 advances across it (`hasDerivAt_charJacobian`). That reweighted kernel is
 481 again even in the separation, so the same characters diagonalize it and ev-
 482 ery mode's rate is one of its eigenvalues. Where the winding advances by
 483 at most a quarter turn across every separation the coupling reaches, the
 484 reweighted kernel is nonnegative and no rate is positive (`charJacobian_`
 485 `stable_of_quarter_turn`); where some rate is positive, that mode grows
 486 without bound (`charJacobian_growing_mode`). Moving coupling mass to
 487 longer separations is what carries a phase advance past a quarter turn, and
 488 that is the mechanism the decay-length sweep varies. The criterion bounds
 489 rates rather than trajectories: asymptotic stability is not proved, so the
 490 states that persist occupy a narrower range than it admits, and the bound-
 491 ary quoted above is measured and not derived from it.

492 The same reweighted kernel measures the potential's curvature. A wind-
 493 ing has $r = 0$ where a global minimizer of the Lyapunov potential has $r = 1$,
 494 so it is not one (`char_not_potential_min`): the quarter-turn condition on ini-
 495 tial data does not merely decline to cover a winding, it must exclude it. Along
 496 a mode, the potential's second variation is that kernel's quadratic form, so a
 497 growing mode is a direction of falling potential and a winding carrying one
 498 is not a local minimum either (`char_not_local_min_of_charLambda_pos`).

499 An amplitude layer decides which of these states exist, and leaves what

500 the resultant reports unchanged. A phase state carries no amplitude. Ev-
 501 ery oscillator sits on the unit circle, so a field with an amplitude node has
 502 no representation in these dynamics. The real Stuart–Landau extension
 503 supplies one on the same coupling, and leaves intact the gradient struc-
 504 ture the convergence results rest on. The state becomes $z_i \in \mathbb{C}$ with $\dot{z}_i =$
 505 $(\mu - |z_i|^2)z_i + \sum_j A_{ij}(z_j - z_i)$. Its phase equation is the Kuramoto equation
 506 with each coupling weighted by a ratio of amplitudes (`amplitude_phase_is_`
 507 `kuramoto`). The phase model is this one with the amplitude ratios set to
 508 one, rather than a separate object.

509 What the added degree of freedom decides is which windings exist at all.
 510 On a periodic array of any dimension, this sheet included, a winding requires
 511 $\mu + \lambda > 0$. Here $\lambda = \sum_d f_d(\cos \phi_d - 1)$ is the eigenvalue the kernel supplies,
 512 summed over separations, with ϕ_d the phase the winding advances across d
 513 (`exists_charState_iff`; on the ring $\phi_d = 2\pi qd/n$). For the nonnegative
 514 kernels used here $\lambda \leq 0$, and moving coupling mass to longer separations
 515 lowers it further. A broad enough kernel therefore removes the winding out-
 516 right. The sweep above reaches that same boundary by a different route:
 517 the phase model retains every winding at every coupling, and loses this one
 518 under integration instead. The two layers bound the same regime from op-
 519 posite sides. A phase recording does not separate a state the amplitude layer
 520 excludes from one the dynamics fails to sustain.

521 These states are windings and not spirals, and on the sheet that is a
 522 theorem rather than a shortage of dimensions. A cortical phase singularity

523 is a point at which the amplitude vanishes and around which the phase turns
 524 once, read in practice as the circulation around a small plaquette. The states
 525 above have equal modulus at every site, so that circulation is exactly zero
 526 on every plaquette (`plaquetteCirculation_eq_zero_of_winding`). Nor can
 527 the layer host a single defect. Over a periodic sheet the circulations of *any*
 528 phase field sum to zero, so defects arrive in cancelling pairs (`no_isolated_`
 529 `defect`). The sweep’s near-zero measured defect density agrees with the first
 530 of these and says nothing beyond it.

531 Two limits bound the layer for a reader of the cortical literature. Relax-
 532 ational dynamics has no propagation speed, which is the quantity a cortical
 533 wave measurement leads with. And $|z|$ is a normal-form radius, which a mea-
 534 sured envelope is not; this work identifies the two nowhere. Supplemental
 535 Material gives the sweep, its uniform control, the amplitude layer and the
 536 standing wave it admits inside the band, together with the remaining limits
 537 of both.

538 A vanishing resultant is the absence of global agreement and not of agree-
 539 ment. Eq. (6) read on a nearest-neighbour patch of a winding gives a dis-
 540 crepancy bound of $2\sqrt{2}L|\sin\phi|$, with ϕ the phase advanced across the patch,
 541 which falls to zero as the same winding is spread over more sites (`chord_le_`
 542 `of_char_patch`), while the global form of the same bound is vacuous at $r = 0$.
 543 Encoders shared on overlaps therefore still bound content disagreement in-
 544 side a patch of a state that no global observable calls coherent. The winding
 545 number itself stays unconnected to content.

546 This qualifies what a phase trace can establish. An onset criterion stated
547 in r alone registers a change of basin as readily as a change of coupling.
548 The protocol of Section 7.3 therefore needs a spatial observable beside the
549 resultant.

550 **6.4 Coherent adaptation can lose environmental align-** 551 **ment**

552 An adaptive coupling rule might supply representational structure, but its
553 objective must reward the relevant agreement. For symmetric coupling, pair-
554 wise sine terms cancel in the population mean drift. Minimizing squared drift
555 at that fixed mean therefore rewards equalizing drift across nodes. In the
556 frequency-cluster control, this favors coupling between unlike-frequency clus-
557 ters, whereas the diagnostic environmental template rewards coupling within
558 clusters.

559 The formalization proves descent of the squared-drift objective for a spec-
560 ified gradient flow at fixed phases. Its functional differs from thermodynamic
561 entropy production: at a coherent stationary density the probability current
562 vanishes while phase-averaged squared drift is positive. Supplemental Mate-
563 rial gives the operator, its gradient and the conditions of the descent result.

564 The finite control co-evolves noisy phases and symmetric, nonnegative
565 couplings at fixed total resource. Its template is a diagnostic covariance of
566 latent signals, never a training input. Across three seeds, second-half order

remains between 0.9167 and 0.9222, while the within-cluster to between-cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–0.6135 (Fig. 5). Random-update, permuted-gradient and frozen-kernel controls are reported in Supplemental Material.

The immediate post-update objective reduction disappears when averaged over complete update intervals. A prespecified grid of 6 rate–cadence combinations contains no case meeting the 5% sustained-reduction criterion while retaining coherence and losing alignment. The finding is coherent adaptation away from the specified structure, with no demonstrated sustained cost reduction.

Kernel alignment and overlap compatibility are different observables. This control addresses acquisition of its designated template under its particular objective. Learning models can explicitly include data in an objective (Corberi et al., 2026); an account of agreement about the cup must specify the content its learning rule is meant to preserve.

6.5 Content agreement needs independent decoders

A neural test needs independently identified domains, local descriptions and restriction maps. Two decoders should estimate a shared content on overlapping domains before their agreement is examined. Discrepancy requires a null under decoding noise and controls using deliberately compatible and incompatible examples. An implementation on constructed data separates these cases at $AUC = 0.986$, on a 95% interval of $[0.980, 0.992]$ resampled by

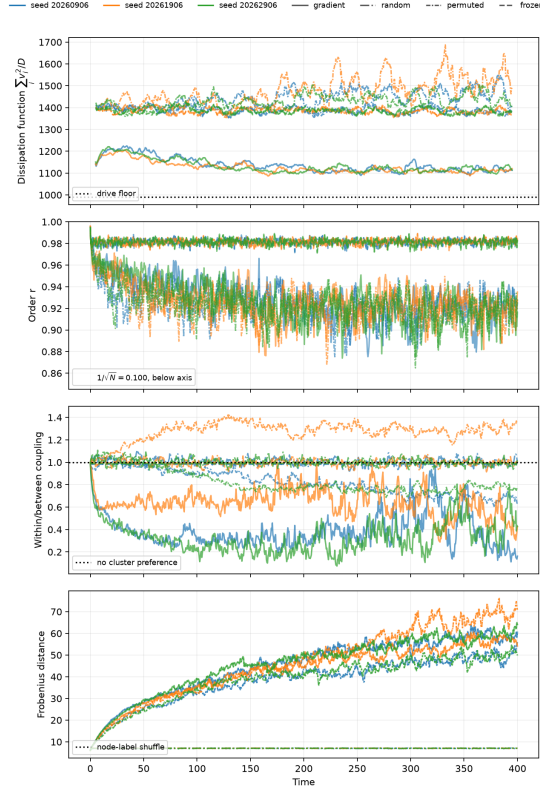


Figure 5: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), permuted gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

trial; the compatible construction scores 0.476 on an interval containing one half. Shared-prior shrinkage can nevertheless manufacture agreement, and regional bias can manufacture disagreement. Per-patch decoding diagnostics detect these failures; the compatibility statistic alone does not.

Phase agreement cannot replace that measurement. On constructed content, phase carries a share 1.000 of the content entropy when content is built from it and 0.000 when it is not. A phase-derived statistic reports compatibility on phase-locked patches whose local descriptions disagree. The implemented content statistic detects their incompatibility. These controls validate a measurement on known constructions; a neural content model and decoder remain to be established.

Choosing the cover also chooses a candidate subject: trivial descriptions can agree while carrying little content, and a partition can conceal disagreement. The useful choice must therefore be constrained by what neural activity encodes. The exact composition assumes compatibility of its supplied cover. The content mechanisms above identify additional hypotheses under which agreement can arise; they leave the empirical choice of content and the connection to that cover explicit.

6.6 What an informative phase-density measurement requires

The concentration range and the dependence structure decide whether the concentration–coherence relation can be tested at all. Under the specified log-density estimator and bias-floor criterion, discrimination at the observed concentration ceiling $a = 0.542$ requires 3000 independent sites in the scanned designs, rising to 31000 when sites share a mean in groups of 10. These counts describe that estimator and dependence model; the intended recording requires its own calibration. Supplemental Material reports the bin-count controls and the full dependence-aware design.

That ceiling comes from exercising the estimators on public propofol-sedation EEG (Bajwa et al., 2025), pooling channels in 100-ms windows. Bipolar and common-reference montages produce different residuals. The bipolar results are compatible with the Bessel relation over a concentration range too narrow to distinguish it from its linear approximation at the observed scatter, which is the range the design scan takes as given. Temporal pooling changes the estimand, and sedation recordings leave the awakening hypothesis untested.

An informative dataset therefore needs sufficient concentration range or precision, calibrated estimators and an observation model for dependence, source mixing and reference choice. Public mouse calcium imaging provides thousands of spatially located neurons across waking, sleep and anesthe-

sia (Oomoto et al., 2026), but its 7.65 Hz frame rate is below the phase-
estimation band used here. Site count, state coverage and bandwidth are
separate requirements.

7 The cortical hypothesis and an awakening test

The cortical proposal identifies a physical source of coupling and asks whether
its changes help explain recovery. The tests in Section 6 determine what
observations must distinguish before a phase trace can support this identifi-
cation.

7.1 Physical identification and calibration

The cortical hypothesis identifies part of effective coupling with endogenous
extracellular electromagnetic interactions. Fields can modulate spike timing
(Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
propose coordination of distributed neural information (Pinotsis & Miller,
2022; McFadden, 2020). Phase organization under propofol motivates study-
ing this regime (Bastos et al., 2021), but does not measure coupling.

The formal field predicate requires a jointly continuous kernel, a normal-
ized substrate, coupling equal to the kernel average, and no single site pro-
viding more than half that strength. Toy constructions satisfy it. A cortical

648 realization must specify the scale at which the kernel and its reduction ap-
 649 ply: calcium imaging reports different state-dependent modularity patterns
 650 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

651 The spatial reduction is also testable. A row-sum aggregation at zero
 652 diagonal tracks the uniform-coupling control over the tested spatial decay
 653 range, while frequency heterogeneity displaces the operational threshold. The
 654 fit window biases both estimates relative to $2D$; agreement with the control
 655 is therefore the supported conclusion. Supplemental Material reports the
 656 reduction, its numerical error and the distinct assumptions needed for the
 657 kernel-average identification in Eq. (5).

658 The field commitment requires physical evidence beyond a choice of math-
 659 ematical representation. A finite matrix spread over positive-mass cells pre-
 660 serves its total coupling and phase correlation exactly. Point-supported
 661 vanishing under an atomless measure and a disadvantage from excluding
 662 a better-correlated pair concern specified support constraints. Neither com-
 663 parison supplies energy prices or a biological–digital distinction.

664 A spike-time shift alone supplies no coupling rate. The dimensionless
 665 product of population size N , timing sensitivity s , field amplitude E and
 666 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N s E f. \quad (15)$$

667 This rate describes how induced timing shifts accumulate into drift. Nei-

ther it nor the installation factor κ follows from a heat bound. Testing the physical threshold requires independent estimates of the interaction kernel, its aggregation and diffusion in the same cortical regime.

7.2 Extracellular geometry and recovery

The proposed control variable is extracellular geometry. Extracellular volume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková & Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density, an effective-medium description predicts that a larger extracellular fraction raises conductivity and reduces field amplitude.

The coupling-gated hypothesis consequently predicts that extracellular contraction on awakening raises E and, holding the other factors in Eq. (15) fixed, raises effective coupling. This argument supplies a direction; the geometry–coupling relation, its magnitude and its role in human recovery require measurement.

For a smooth crossing at nonzero speed, evaluating the stationary branch along the changing control gives the benchmark

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (16)$$

This follows from Eq. (3) at fixed D and concerns the stationary branch

687 evaluated at $K(t)$. Reading it as observed recovery requires an indepen-
688 dently justified tracking approximation. The crossing controls in Section 6.2
689 leave that approximation unsupported at onset, even though a separation of
690 geometry and phase time scales can support it away from threshold. The
691 awakening hypothesis therefore needs a calibrated dynamical prediction for
692 an operational onset above the finite-size fluctuation floor. Crossing K_c alone
693 does not supply a recovery latency.

694 **7.3 A protocol for the awakening experiment**

695 The physical test has six requirements:

- 696 1. **Coupling and diffusion in the same units.** Estimate them from re-
697 solved phase dynamics, separating heterogeneity and observation noise.
698 Measure the drive or perturb the system to distinguish endogenous cou-
699 pling.
- 700 2. **Geometry calibrated to coupling.** Validate the geometry observ-
701 able and its relation to effective coupling, with uncertainty. The con-
702 ductivity argument supplies only the expected direction.
- 703 3. **Calibrated phase measurements.** Estimate r by resultant length
704 and a by unweighted regression of log density on centered cosine.
705 Choose site count and binning from a dependence-aware calibration
706 over the intended concentration range. The tested designs require

707 thousands of sites at the observed concentrations; temporal pooling
708 changes the spatial estimand.

709 4. **An independent recovery endpoint.** Measure behavioral recovery
710 separately from phase organization.

711 5. **A predeclared dynamical onset criterion.** Set a coherence thresh-
712 old above the finite-size floor and evaluate its latency under the mea-
713 sured coupling trajectory, including ramp speed or fluctuations. Record
714 a spatial observable beside the resultant, since a winding field reports
715 a low resultant at coupling that sustains local order (Section 6.3).

716 6. **Matched model comparison.** Compare coupling increase, diffusion
717 change, common drive and alternative recovery models under the same
718 observation and filtering assumptions.

719 Failure of the concentration–coherence relation over an informative range
720 constrains the density model. Failure of the calibrated geometry–coupling
721 association constrains the physical gate. Better prediction of independent
722 recovery by a competing mechanism counts against its proposed role. A
723 favorable outcome still leaves the content of coordinated activity to be ex-
724 plained.

725 8 Discussion

726 The account set out in Section 1 is conditional at every step: each connection
727 in it is a hypothesis supplied rather than a result derived, and Table 1 names
728 all eight. What follows takes those conditions in turn, as commitments the
729 proposal incurs, as measurements that would bear on them, and as questions
730 the framework leaves open.

731 8.1 Unity and a minimal self as empirical commit- 732 ments

733 We propose the state glued from compatible local descriptions as a corre-
734 late of conscious unity, and its accurate internal reconstruction as a correlate
735 of a minimal self. The motivations are that different contents belong to
736 one experienced situation and that the situation is experienced from a per-
737 spective. The conditional results identify relations a realization would need:
738 sufficient installed resources for the specified phase model, agreement about
739 shared content, and an encoding that supports reconstruction at the required
740 fidelity.

741 Compatibility allows different descriptions to contribute different infor-
742 mation. It also permits uncertainty and conflicting judgments: apparent cup
743 position and judged cup position can be distinct quantities in a consistent
744 description of their discrepancy. Which quantities are shared must be de-
745 termined independently. Choosing them solely to secure agreement would

746 deprive the proposal of empirical force.

747 The minimal-self proposal concerns the perceiver’s position and condition
748 within the situation, such as the cup’s location relative to a body that can
749 reach it. The encoding could be implicit and need carry no autobiograph-
750 ical narrative. Its proposed necessity is philosophical: even an unreflective
751 episode has a perspective. Gluing by itself requires no self-encoding, and re-
752 constructability alone supplies no functional role. Altering the encoded hand
753 or viewpoint while holding the external scene fixed should change estimates
754 of the perceiver’s relation to the cup as the readout predicts. Whether that
755 represented perspective is experienced remains a further question.

756 **8.2 What a test of the full proposal must establish**

757 Both uniqueness results concern specified objects. They choose neither the
758 cover nor the encoding, and do not select between candidate subjects. In-
759 tegrated information theory treats exclusion explicitly (Tononi et al., 2016);
760 here selection requires independent evidence about neural organization. A
761 constant readout can contract and fix a predetermined state while discard-
762 ing its input. The family criterion in Section 4.2 tests the sensitivity that
763 this construction lacks, provided the relevant states and fidelity are declared
764 independently.

765 A test must separately measure coupling relative to diffusion, content
766 compatibility and self-reconstruction, alongside an independent endpoint for
767 experience. Meeting all three without a unified episode would challenge suffi-

768 ciency. A unified episode with incompatible descriptions would challenge the
769 unity identification; one lacking the specified self-encoding would challenge
770 the reflexive requirement. Awakening, dreaming and dissociative states offer
771 possible settings, but absence of report cannot by itself define absence of
772 experience.

773 The next physical requirement is a calibrated geometry–coupling relation
774 and resolved phase dynamics. The installation bound contributes a nec-
775 essary resource condition where its hardware representation applies; cortical
776 mode prices and field-mediated coupling require measurement. For the repre-
777 sentational connection, an informative experiment asks whether a controlled
778 change in coordination alters independently decoded content agreement, with
779 controls for sensory information and decoder quality. The constructed-data
780 estimator establishes feasibility for specified contents. Neural local descrip-
781 tions, shared overlap encoders and the self-encoding have yet to be identified.
782 The population-size dependence of the phase-to-content bound further limits
783 its quantitative use.

784 The separation results explain why these measurements must be inde-
785 pendent. A stationary distribution leaves several recovery mechanisms indis-
786 tinguishable. Equal phase order can accompany incompatible descriptions
787 or different task information. The tested plasticity rule preserves coherence
788 while losing its designated template. Each finding concerns the specified
789 model and observable; their role is to delimit the evidence needed for the
790 proposed connections.

791 8.3 GPUs and the limits of no-go claims

792 For a large language model (LLM) running on graphics processing units
793 (GPUs), the framework requires identifying a physical system whose local
794 descriptions form one state and whose internal encoding reconstructs that
795 same state. The coupling-gated route E56–E78 requires a calibrated physical
796 coupling, its noise regime and a connection from coordination to content
797 agreement. An attention matrix or a shared clock supplies none of these identifications on its own. Computing a field model likewise leaves its realization
798 in the computing substrate to be established. These physical obligations also
799 apply to the cortical proposal.

801 Unity requires the candidate’s local descriptions to agree on indepen-
802 dently identified shared quantities (Section 3). Common timing and fluent
803 output do not establish that agreement. Digital representations can sat-
804 isfy compatibility, however, so finite hardware is compatible with the gluing
805 construction. The empirical commitment is to identify the relevant cover
806 and contents with one experienced situation; mathematical uniqueness alone
807 selects no subject.

808 For self, the system must carry an encoding $E(s)$ and an internal readout
809 R of the same global state. E89 requires $F = R \circ E$ to contract and fix
810 the state produced by gluing. A constant readout shows why a unique fixed
811 point alone is insufficient for informative self-representation. Reading that
812 encoding as the restriction of the state to a region of the machine puts the
813 demand on the region, which has to resolve the relevant family before any

814 readout is chosen. The additional family criterion in Section 4.2 supplies
815 a test: if two relevant macrostates satisfy $E(s) = E(t)$ and $d(s, t) > 2\varepsilon$,
816 Eq. (10) rules out reconstructing both within error ε . An intervention outside
817 the encoding region’s causal past gives a corresponding obstruction before
818 a declared communication deadline. Identical verbal reports alone do not
819 establish identical internal codes. These conditional no-go results exclude
820 a specified reconstruction at a specified fidelity or deadline. They apply to
821 biological and digital candidates alike.

822 The hardware comparisons in Supplemental Material have similarly re-
823 stricted scope. Fixed wiring can impose a correlation disadvantage, and
824 point-supported kernels vanish under an atomless continuum measure. Nei-
825 ther comparison makes the representation conditions unattainable on GPUs.
826 The installed-energy condition of Section 2.2 is restricted in the same way.
827 Once a candidate’s installed energy stands far above the thermal floor, which
828 it does for any macroscopic substrate, the condition constrains that candidate
829 only through κ , the conversion from installed energy to coupling; κ is hard-
830 ware data supplied by mode profiles and prices, and no result here derives
831 it for cortex or for a processor. Supplemental Material reads the condition
832 against a cortical metabolic budget and returns the conversion factor it would
833 require rather than a verdict. A universal exclusion would need a necessary
834 condition for consciousness and a proof that every relevant implementation
835 violates it. The present conditional composition supplies neither.

836 Whether experience requires a living substrate is the explicit subject of

837 biological naturalism (Seth, 2026a,b). On that view the properties ground-
 838 ing experience are those of self-sustaining organisms, and simulating them
 839 does not secure them. The conditions stated here name no biological mecha-
 840 nism. Computation described as such does not satisfy them either: a candi-
 841 date must exhibit a physically realized coupling in a measured noise regime,
 842 agreement about independently identified shared quantities, and an internal
 843 reconstruction meeting a declared fidelity. Whether a realization of those
 844 conditions that is not alive would be accompanied by experience is the point
 845 at issue, and these results do not settle it. A related position locates the
 846 difference in the mode of computation rather than in life as such: biologi-
 847 cal systems compute across scales that do not separate, and alongside dis-
 848 crete operations carry continuous-valued dynamics belonging to their phys-
 849 ical medium (Milinkovic & Aru, 2026). The coupling condition stated here
 850 is one measurable instance of that requirement, fixed to a physical field in a
 851 declared noise regime, and a description of such a field does not satisfy it.

852 The neuroscientific case against *human-like* consciousness in text-only
 853 LLMs concerns embodied sensory interaction, thalamocortical organization
 854 and biological regulation linking perception and action to survival (Aru et
 855 al., 2023). Their absence supports skepticism, conditional on their causal
 856 relevance to experience. Those mechanisms are not requirements of the ab-
 857 stract gluing and reconstruction definitions. Workspace-like representations
 858 reported in some LLMs (Gurnee et al., 2026) provide candidate organization
 859 to investigate. Within this framework, the relevant evidence would identify

860 compatible contents and an informative internal reconstruction, or demon-
861 strate their failure under controlled interventions.

862 **8.4 Explanatory scope**

863 Integrated-information, global-workspace and predictive-processing accounts
864 organize unity, access and modelling differently (Tononi et al., 2016; Baars,
865 1997; Dehaene et al., 2014; Friston, 2010). The present proposal contributes a
866 candidate physical interaction and conditional relations between local content
867 and self-encoding. Gluing and reconstruction permit multiple substrates;
868 the electromagnetic commitment enters through physical calibration. The
869 formalization checks implications under their hypotheses, and its witnesses
870 establish realizability on declared models. They do not together supply a
871 cortical agent.

872 The explanatory question is whether the specified relations occur in neural
873 activity and matter for experience. In the cup example, this means establish-
874 ing how coordination helps maintain one represented situation and how the
875 perceiver’s place within it contributes to the episode. The proposal identi-
876 fies candidate correlates and tests of that identification. Why those relations
877 would be accompanied by experience remains unexplained.

878 **Data and code availability**

879 The Lean development, simulation scripts, tests and saved numerical sum-
880 maries are maintained in the GitHub repository `bobaseb/coupling_gated_`
881 `cortical_coherence`. Publication simulation values are generated from
882 saved summaries without rerunning production sweeps. The exploratory
883 EEG analysis uses OpenNeuro dataset ds005620 (Bajwa et al., 2025); pre-
884 processing and selection are described in Supplemental Material. The repos-
885 itory contains the reproducible analysis and does not redistribute the raw
886 recordings.

887 **Funding and competing interests**

888 This research received no external funding. The author declares no compet-
889 ing interests.

890 **Use of AI tools**

891 AI assistance to this work was substantial and is not separable by task. Sev-
892 eral coding assistants and chat interfaces were used in an interleaved way
893 throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,
894 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra; GPT-5.6 Terra,
895 Luna and Sol), Claude Code (Claude Opus 5), Gemini Pro through its web in-
896 terface, and Anthropic’s Fable and DeepSeek v4 Flash through OpenRouter,

897 accessed with the Hermes CLI. Between them they contributed to the Lean
898 development, the simulation and analysis code, and the drafting and revision
899 of both documents. Because the tools were used in combination and in alter-
900 nation, no section, proof, figure or result is attributable to any one of them,
901 and none is claimed to be. The author is responsible for all content, including
902 everything a tool produced. Machine checking establishes the stated Lean
903 results under their hypotheses; it does not validate biological interpretations
904 or replace scientific review of generated code and prose.

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